HABIT PROGRESSION AND REGRESSION—A LABORATORY STUDY OF SOME FACTORS RELEVANT TO HUMAN SOCIALIZATION

JOHN W. M. WHITING

Yale University

AND

O. H. MOWRER

Harvard University

Received August 12, 1943

There are many problems in human socialization which involve the principles of learning. One of these concerns the relationship between habits which supplant one another during the life cycle. A number of anthropologists and sociologists (1, 9, 11, 19, 24) have pointed out that there are numerous forms of behavior which are considered right and proper during one stage of the life cycle but wrong during another stage. For example, in our culture, an infant cries when he is hungry. Later, when he reaches the status of child, this habit is either punished or extinguished; and asking for food supplants it. Still later, when he becomes an adult, he must work for food rather than expect to get it for the asking. Despite the obviously complicated nature of this example, the fact remains that a succession of habits is attached to the hunger drive. A similar series is attached to other drives as well. Infants may eliminate as soon as there is sufficient bladder or colon tension. Children may not do so, but must wait until they have gone to the toilet. Children may physically attack their competitors in the play group; adults, in general, may express only verbal aggression and that only under certain conditions. Thus socialization frequently involves habit progression. It is therefore important to analyze the underlying principles and to determine the way in which they operate.

Habits which have been superimposed upon other habits stand in sharp contrast to those which have been continuously rewarded throughout the life cycle. In the former case, the more recently acquired alternative habit must always compete against an older habit; *i.e.*, there is a constant tendency toward regression. On the other hand, when progression is not involved, there is no such competing tendency and the habit is therefore more stable. Clinical observers in our society have frequently commented on the emotional stress which is engendered by the presence of these regressive tendencies. We have not attempted to study the emotional aspects of such conflicts, but we have explored the habit dynamics involved in situations where there is habit progression and a resulting tendency toward regression.

A number of prior animal studies have dealt with the dynamics of habit progression and regression (5, 8, 10, 12, 17, 18, 20, 21, 22) but none of these has systematically investigated the two problems with which we are here concerned. All workers in this field have recognized that in order to produce habit progression, *i.e.*, in order to induce an organism to give up an originally preferred, or

"fixated", adjustment and acquire a new one, it is necessary for the original adjustment to be interfered with in some manner. Various methods of achieving this end have been employed, but these have not been compared in any orderly fashion. Such a comparison is, however, important from two points of view: 1) the ease with which habit progression can be produced, and 2) the likelihood of later regression. In this study one of our two major aims has been, therefore, to compare, both with respect to habit progression and subsequent regression, three methods of interfering with the original habit, *viz.*, the use of non-reward, a physical barrier, and active punishment.

Various incidental observations have suggested that both habit progression and the likelihood of later regression are also influenced by the relative effortfulness of the original and alternate adjustments. If the alternate habit is less effortful than the original one, progression can be easily produced and the chances of regression are relatively slight. On the other hand, if the alternate habit is more effortful than the original habit, progression is comparatively difficult to produce and the chances of regression are proportionately increased. From these facts, it seemed probable that in those situations in which the new adjustment is more arduous than the original one, the difficulty of producing progression and later susceptibility to regression would be positively related to the magnitude of the discrepancy between the degree of effort involved in the two adjustments. With this second problem in mind, we have accordingly employed a dynamic learning situation in which the factor of effort could be controlled and systematically varied, along with the three methods of producing habit progression which have already been mentioned.

APPARATUS

The general plan of the elevated maze used in this investigation is shown in Figure 1. Placed at the starting point, S, a rat was able to go to the goal, G, by either of two paths. What we shall call the original path, S–O–G, was approximately 5 feet long. Either of three alternate paths could be made available: S–A_S–G, the shortest alternate path, which was 10 feet long; S–A_I–G, the intermediate alternate path, which was 20 feet long; and S–A_L–G, the longest alternate path, which was 40 feet long. All paths were 5 inches wide, with a ½ inch railing on either side. The extent of the elevation above the floor was 3 feet.

PROCEDURE

In all, 45 young adult albino rats were used in this experiment. At the outset they were randomly divided[1] into 9 groups of 5 animals each. Each group was then randomly assigned to one cell of the diagram shown in Figure 2.[2] Since all 45 animals could not be run concurrently in this experiment, they were run in sets of 9 animals, each set consisting of one animal randomly selected from each of the 9 cells. However, in the following account of the various steps in the

[1] With one exception having to do with the distribution of male and female animals (see Appendix).

[2] For a discussion of the logic according to which this experiment was designed, see Fisher (2).

procedure, we shall, for the sake of simplicity, speak as if all 45 animals were run concurrently.

Part I—Fixation on the original path. For a period of one hour on each of the two days preceding the onset of regular training, all animals were allowed to

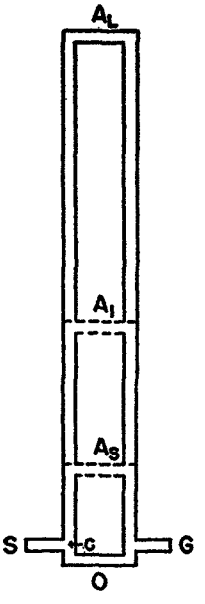


FIG. 1. DIAGRAM OF THE ELEVATED MAZE (VIEWED FROM ABOVE) WHICH WAS EMPLOYED IN THIS INVESTIGATION

The removable bridges, represented by the dotted lines, were so constructed as to bar entrance to the parts of the maze lying beyond them.

LENGTH OF ALTERNATIVE PATH	METHOD OF INTERFERING WITH ORIGINAL HABIT		
	Non-reward	Barrier	Punishment
Short	SN 5	SB 5	SP 5
Intermediate	IN 5	IB 5	IP 5
Long	LN 5	LB 5	LP 5

FIG. 2. DIAGRAM SHOWING THE NINE DIFFERENT EXPERIMENTAL CONDITIONS USED IN THIS INVESTIGATION

SN stands for Short-Non-Reward; SB for Short-Barrier; etc. Five animals were run under each of the nine different conditions.

explore that arrangement of the maze on which they were later to be trained: *i.e.*, for Groups SN, SB, and SP, the alternate path was S–A_S–G; for Groups IN, IB, and IP, the alternate path was S–A_I–G; and for Groups LN, LB, and LP, the alternate path was S–A_L–G. No food was provided at any point in the maze during this preliminary habituation period.

On the first day of regular training each animal was treated as follows: On

trial 1, the animal was placed on the maze at G and given a small standardized pellet of moist food (Purina Dog Chow). On trial 2, the animal was placed on path O about one foot from G and then lured to and fed at G. The same procedure was repeated on trials 3, 4, and 5, with the animal being placed approximately one foot further from G and nearer S on each successive trial. On trials 6–15, the animal was placed at S and allowed to go to G, either by the original path or by the alternate path which was open to it, without luring. On the four following days, each animal was likewise given 15 trials, but on these days the animals were always started at S and were never lured. All animals were fed to satiation after each daily training session. They were then deprived of food until the next experimental session, approximately 23 hours later. By the end of the initial 75 trials, all animals had developed a marked (though not perfect) preference for path O and usually ran by this path from S to G in 2 to 5 seconds. On all trials the experimenter recorded the time taken by each animal to go from S to G, the path chosen, and certain incidental observations.

Part II—Progression to alternate path. Following the 5 days of fixation on path O, all animals were given 30 days of training which was designed to make them abandon path O and accept the alternate path which was available to them. Conditions were now such that the animals in Groups SN, IN, and LN received food at G only if the went from S to G via the alternate path. If they went to G via the original path, they received no food and were immediately returned to S by the experimenter. If, on the first day of this new training, an animal had not tried the alternate path after 150 non-rewarded trips to G via the original path, the animal was lured around the alternate path to G, where it now found food. If, after an animal had once been lured to G via the alternate path, it made as many as 10 successive trips to G via the original path, it was again lured to G via the alternate path. On the first day, four animals in Group SN, two in IN, and three in LN had to be lured to G one or more times. On the second day, one animal in Group IN had to be lured to G once, but this procedure was never necessary with any animal after the second day. During this part of the experiment each animal was required to make 15 trips per day to G via the alternate route.

The animals in Groups SB, IB, and LB were physically barred from going to G via the original path during this part of the experiment by means of a square of window glass (which was probably invisible to the rats at a little distance) placed across this path midway between S and G. For the animals in Groups SP, IP, and LP, it was possible to go to G either by the original or by the alternate path, and in both cases they obtained food; but if they selected the original path they received an electric shock from a grill located flush with the surface of the maze midway between S and G. In both the barrier and the punishment groups, if an animal did not reach the goal within five minutes after being released at S, the experimenter attempted to lure it to the goal via the alternate path.[3] If an

[3] The use of luring in this and later parts of the experiment is reminiscent of the use of "instruction" in the socialization of children. By means of symbols we are able to say to a child, "You can solve your problem by doing thus and so." Unable to *tell* our subjects that food would be available to them if they would only go to G via the alternate route, we had to *show* them.

animal refused to be lured, it was picked up and carried to G, where it was allowed to eat the available pellet. These special procedures were never necessary, however, after the first few days and were resorted to in the beginning only as a means of preventing an occasional refractory animal from taking a disproportionate amount of time to progress to the alternate path.

During this part of the procedure, an "error" for the non-reward animals consisted of going from S to G via O; for the barrier animals it consisted of touching the glass barrier with any part of the body (including the vibrissae); and for the punishment animals it consisted of receiving a shock from the electrified grill. Running time per trial, route followed, behavior at the choice point, and certain other items were also recorded by the experimenter.

Part III—"Spontaneous" regression. The purpose of this part of the experiment was to determine how the tendency to revert to the original path was related to the length of the alternate path and to the methods which had been employed to produce habit progression. Accordingly, all animals were run 15 trials a day for 5 days under objectively the same conditions as prevailed during Part I, *i.e.*, with all "sanctions" against taking path O withdrawn. This meant that the shock was turned off for the punishment groups, that the glass barrier was removed for the barrier groups, and that food was available at G for the non-reward groups, regardless of whether they reached G by path O or by an alternate path. The data recorded were the same as in Part I.

Part IV—Forced regression. Since some of the animals showed comparatively little tendency to regress "spontaneously" during Part III, it was decided to force them to do so. Therefore, on the day following the completion of Part III, all animals were given 15 trials with path O as the only means of reaching G, the alternate path being blocked by a barrier placed just to the left of the choice point. If, on any of these 15 trials, an animal failed to go to G within twenty minutes after being released at S, it was lured to the goal. This practice was resorted to simply as a means of setting a limit to the amount of time which any one animal could consume in making the 15 required runs.

THEORETICAL EXPECTATIONS AND EXPERIMENTAL FINDINGS

From the training of intra-human animals both in the laboratory and in everyday life and from the socialization of children in primitive as well as civilized cultures, a considerable amount of agreement has been reached concerning the basic principles governing the process of learning. These principles generate certain expectations concerning the probable results of the experiment here reported. It is now our purpose to compare these expectations with the results actually obtained.

Part I. As a result of carefully controlled research, as well as common experience, it is known that

1. Of two or more acts which precede a rewarding state of affairs, that act which is temporally nearest the reward is the most strongly reinforced (gradient of reinforcement); and that

2. Of two or more acts which precede a rewarding state of affairs, that act

which involves the least expenditure of effort receives, other things being equal, the greatest amount of *net* reinforcement (law of least effort).

Since the act of turning to the right at the choice point in the maze used in this investigation tended to be more quickly followed by reward than did the act of turning to the left at this point, the first of these principles demands that right-turning at the choice point should be more strongly reinforced than left-turning.[4] And since running from S to G via the original route required less effort than did the running of the alternate routes, the second of these principles demands that the original path should again tend to become dominant, or preferred.[5]

The double expectation thus generated that O would in all cases become the preferred means of going from S to G is confirmed by the finding that in the the course of the 75 trials constituting Part I of this experiment, the 15 animals for which the short alternate path was available chose path O on 92.0% of the trials, the 15 animals for which the intermediate alternate path was available chose path O on 95.3% of the trials, and the 15 animals for which the long alternate path was available chose path O on 97.7% of the trials. Since it was possible for the rats, by taking the original path, to reach the goal both more quickly and with less effort, it was obvious that a preference for the original path would develop.[6]

In the original statement of our reason for using alternate paths of different lengths, we mentioned only the variation in effort which this arrangement would provide. It is clear, however, that with motivation held constant, more *time*, as well as effort, is required by a rat in traversing the alternate paths than in traversing the original path. In other types of experimental situations, the effects of effort and time can be separated, but in the maze here employed this could not be done without unduly complicating the procedure. Henceforth, when we speak of the effects of the greater length of the alternate paths, it will be understood that we include the differential reinforcement introduced by the time factor as well as by the effort factor.

The principles which led to the prediction that path O would be preferred to the alternate paths also led to the prediction that this preference would not be an all-or-none affair but would vary with the magnitude of the time-effort discrepancy between the original and alternate paths. This prediction is confirmed by the finding that as the alternate path is made longer the preference for the original path becomes progressively stronger: 92.0%, 95.3%, and 97.7%, respectively, for the short-, intermediate-, and long-path groups. When the results for this part of the experiment are analyzed by days, the curves shown in

[4] For a more detailed analysis of the role of the gradient of reinforcement in determining preference for one of a number of different routes to the same goal, see Hull (6).

[5] For discussions of the law of least effort, see Hull (7), Mowrer and Jones (14), and Wheeler (23).

[6] Yoshioka (25) has shown that in order for rats to develop a reliable preference for one of two paths leading to the same goal, one path must exceed the other in length by at least 10%. In the present experiment, the differential between the original and the alternate paths was in all cases much greater than this.

Figure 3 are obtained. These curves show the average number of times during 15 trials that the original path was chosen on successive days by the animals in each group in preference to the alternate path which was available. In addition to the over-all differences between groups, it will be noted that there was a tendency in all groups for the preference for path O to become progressively stronger as training proceeded.

The difference between the short- and the long-path groups is highly reliable, there being less than one chance in 1,000 that such a difference would have been

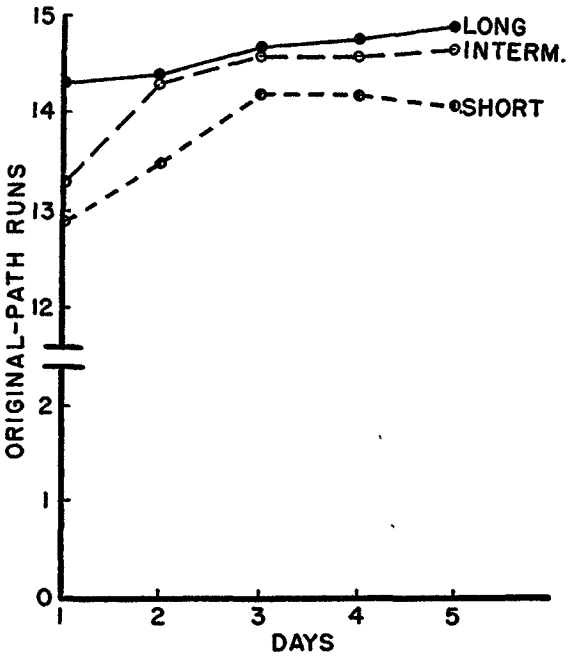


FIG. 3. CURVES SHOWING INCREASE IN PREFERENCE FOR PATH O DURING PART-I TRAINING
Each point represents the average number of path-O choices made per day by the 15 animals which were run under each of the three experimental conditions, *i.e.*, with the long intermediate, and short alternate paths available. Complete preference for path O would have been indicated by a score of 15.

obtained if the "null hypothesis" were valid, *i.e.*, if the length of the alternate paths was of no significance in determining the results. The difference between the long and intermediate and the intermediate and short path groups are significant at the .04 and .05 levels respectively.

The three curves presented in Figure 3 require two further comments. It will be noted that even on the first of the five days of training, the three groups, while clearly differentiated, all showed a decided preference for the original path. This fact is due, not only to the influence of the greater time and effort involved in running the alternate paths, but also to the circumstance that on the first six trials on the first day all animals were lured to the goal *via the original path.* This training necessarily established a bias in favor of path O, over and beyond

the influence of the effort-time discrepancy, but this bias was deliberate. In the first place, the luring was used as a practical device for getting the animals accustomed to running the maze as quickly as possible, without taking the exhorbitant lengths of time frequently consumed on the first few trials of maze learning. Moreover, for reasons which will later become evident, it was desirable that during this stage of training all groups should choose path O as frequently and the alternate paths as infrequently as possible. The luring obviously contributed to the end of decreasing the vacillation between paths which would have otherwise occurred on the first few trials, before the effort-time differential became fully effective. Whether the "latent learning" that presumably occurred during the preliminary habituation to the maze reduced the vacillation between paths during the early stages of regular training is an interesting problem, but our data throw no light on it. The initial habituation was used simply as an expedient for dissipating, before regular training started, the timidity which animals commonly show in new situations.[7]

Part II. This part of the experiment involved two major problems: (1) the relation between the ease with which progression from path O to an alternate path could be produced and the amount of effort (and time) required by rats in running the alternate path; and (2) the relation between the ease with which this progression could be produced and the type of "sanction", "tabu", or "discipline" used to interfere with the fixation upon the original path. We shall discuss the former of these problems first.

Let us assume, for the sake of simplicity, that in Part I of this experiment all animals had developed an equally strong and stable fixation upon path O as a means of going from S to G. From the principles which have already been set forth, it follows that if path O were interfered with, the readiness with which an alternate path would be accepted as a substitute would be a function of the amount of net reinforcement provided by the alternate path. In other words, the longer, *i.e.*, the more effortful and the more time-consuming, the alternate path, the less the habit of choosing it would be reinforced each trial and the more slowly would it accumulate strength, or "cathexis". The results shown in Figure 4 confirm this expectation. Here it will be seen that the number of "errors" made during this part of the experiment was consistently greatest for the 15 animals for which the alternate path was longest (40 feet), least for the 15 animals for which the alternate path was shortest (10 feet), and intermediate for the 15 animals for which the alternate path was of intermediate length (20 feet).

The assumption that all animals had had equal practice on the alternate paths at the beginning of the Part-II training is, of course, not strictly accurate. During the Part-I procedure, the animals for which the short alternate path was

[7] Preliminary observations had indicated that if animals were placed on the maze without prior habituation, they showed considerable anxiety. This was first indicated by great cautiousness of movement and excessive urination and defaecation. Later there was a period of feverish exploration, during which the animals ignored food even though they had not eaten for 24–36 hours. These observations suggest that "curiosity" is perhaps more closely related to anxiety than is ordinarily supposed.

available took this path an average of 6.0 out of 75 trials (15 trials per day for 5 days), as compared with average scores of 3.5 and 1.7 for the intermediate- and long-path groups, respectively. In other words, at the beginning of Part II, the short-path group had received an average of 2.5 (6.0–3.5) more reinforcements per animal on the alternate path than the animals of the intermediate

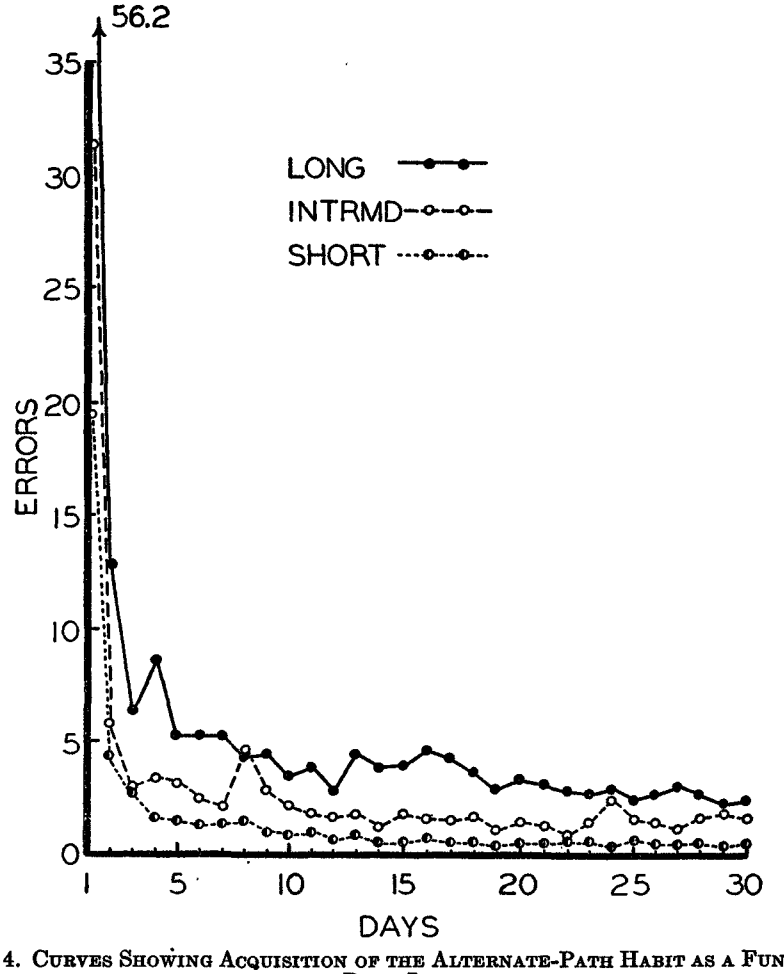


FIG. 4. CURVES SHOWING ACQUISITION OF THE ALTERNATE-PATH HABIT AS A FUNCTION OF PATH LENGTH

Each point represents the average number of "errors" (see text) made per day by each of three groups of animals, during Part-II training.

path group, and the intermediate-path group had received an average of 1.8 (3.5–1.7) more such reinforcements per animal than those of the long-path group. But a glance at Figure 4 shows that the difference between the curves could not be accounted for on the basis of such a small difference in the number of Part-II reinforcements. For example, the animals in the short-path group made fewer errors on the 10th day than the intermediate-path group made on the 30th day,

even though the latter had received 300 (15 per day for 20 days) more reinforcements than the former. It is therefore evident that the differences in the three groups, as shown in Figure 4, can not be accounted for solely in terms of unequal prior reinforcement but are rather mainly the result of the time and effort factors, as stated above.

It was because of this problem that an attempt, previously described, was made to keep the alternate-path reinforcements received by the three groups during Part-I training as nearly equal as possible. Had it been possible to keep the alternate-path reinforcements strictly constant during Part I, the foregoing discussion would have been unnecessary.

Having thus considered the first major problem involved in this part of the experiment, namely, the relation between the length of the alternate path and the ease with which progression from path O to an alternate path could be produced, we may proceed to the second problem, namely, the relation between the ease with which this progression could be produced and the type of "sanction" used to interfere with the fixation upon the original path. It will be remembered that with the onset of the Part-II training, each of the 45 animals used in this investigation was subjected to one of the nine different experimental conditions described in the diagram shown in Figure 2. We have just examined the results which were obtained when the length of the alternate path was varied and the methods of interfering with the original-path habit were counterbalanced so as not to influence the outcome. We now turn to an examination of the results obtained for the three methods of interference, with length of the alternate path balanced out.

Although there is now good reason for believing that the inhibition of habits through "extinction" involves the same basic principles as inhibition through "punishment" (14), a quantitative difference is involved, *i.e.*, the resulting conflict is more severe in the second case. This fact led to the prediction that the animals in this experiment which were shocked for going to G via the original path would abandon this behavior more quickly than would those which were merely refused reward when they took this path. When an animal went from S to G and failed to find food, the only motivational deterrant to the repetition of this act was the fatigue thus created. But when an animal went from S to G and received shock, a considerably stronger conflict was created, involving *both* fatigue and shock: hence the expectation that the "punishment" animals would abandon path O more quickly than the "non-reward" animals. The curves presented in Figure 5 dramatically confirm this expectation.[8]

The magnitude of the difference between the punishment and the non-reward curves would undoubtedly have been less if the intensity of the shock used with the punishment group had been weaker. Since the intensity of the shock could have been varied downward to zero by any desired series of steps, it should, in fact, have been possible to make the punishment curve approximate the non-

[8] See Appendix for a discussion of the statistical reliability of the differences between the curves shown in Figure 4 and in Figure 5.

reward curve to any specified extent. Why the curve for the barrier group shows better learning during the first 21 days than the non-reward group and then crosses the latter curve is a question which is intriguing theoretically; but since any explanation we might offer here would be largely speculative, we shall postpone dealing with this problem until a later time.

One other problem remains to be considered. It will be remembered that we employed three different criteria of "errors" in this part of the experiment. In the case of the punishment animals, an error consisted of their receiving an

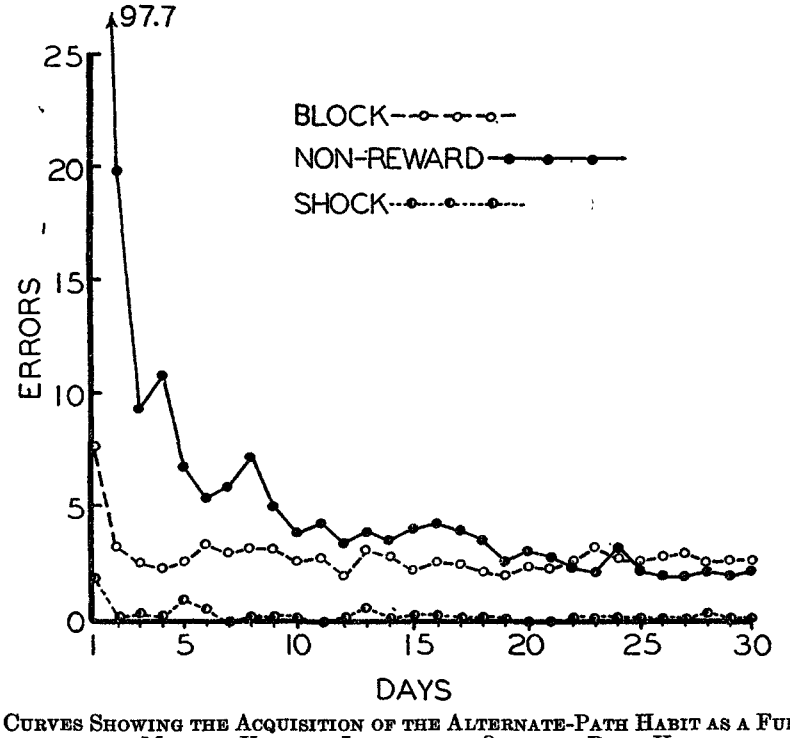


FIG. 5. CURVES SHOWING THE ACQUISITION OF THE ALTERNATE-PATH HABIT AS A FUNCTION OF THE METHOD USED TO INHIBIT THE ORIGINAL-PATH HABIT.

Each point represents the average number of "errors" made by each of the three groups of animals, during the Part-II training.

electric shock; in the case of the barrier animals, it consisted of their touching the glass barrier; and in the case of the non-reward animals, it consisted of their running from S to G via path O. Lest the differences in the scores obtained for the groups which were run under the conditions of punishment, barrier, and non-reward be due merely to the difference in the criteria of errors, rather than to the conditions of the experiment, it seemed wise also to analyze the results in terms of some criterion of performance which would be common to all groups. In order to make this type of comparison possible, we kept a record, not only of "errors" as previously defined, but also of "partial errors", a "partial error"

consisting of any entry into path O at choice point C. In this way we secured a criterion of performance which was exactly the same for all groups.

When the result of the training procedures carried out in Part II are analyzed in terms of "partial errors", the curves shown in Figures 6 and 7 are obtained.

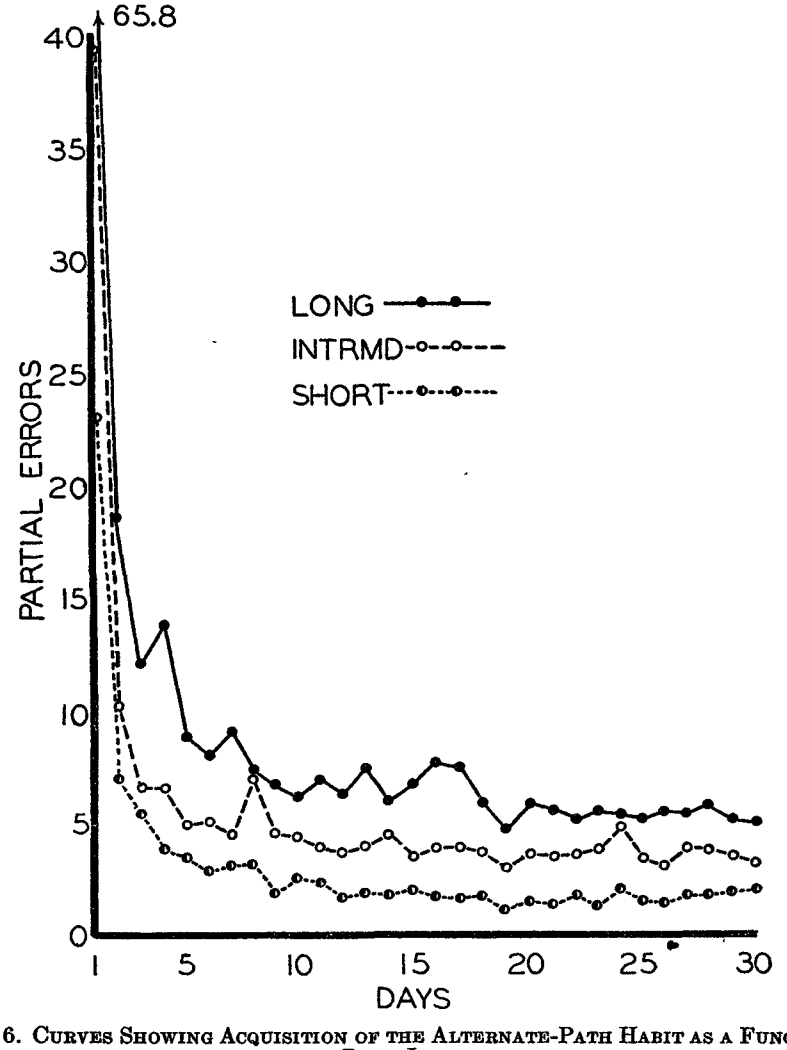


FIG. 6. CURVES SHOWING ACQUISITION OF THE ALTERNATE-PATH HABIT AS A FUNCTION OF PATH LENGTH

Each point represents the average number of "partial errors" (see text) made per day by each of three groups of animals, during Part-II training (cf. Fig. 4).

That all groups made somewhat more "partial errors" than "errors" is due to the fact that every "error" necessarily involved a "partial error" (entry into path O), whereas an animal could make a "partial error" without making an "error" (getting shocked, touching the barrier, or going all the way to G and receiving no reward). The differentiation of the long-, intermediate-, and

short-path groups is even more pronounced in Figure 6 than in Figure 4.[9] However, the separation of the curves for the block, non-reward, and shock groups is much less marked in Figure 7 than in Figure 5. Any attempt to explain why the curves shown in Figure 5 are clearly differentiated and those in Figure 7 are not would at present consist largely of conjecture. The comparison of "errors" and "partial errors" has, however, served to show that the "results" of this part of the experiment *are* different, depending on which measure is used. But, as the following pages will indicate, this finding can not be taken as proof that the differences in the "error" curves shown in Figure 5 are spurious.

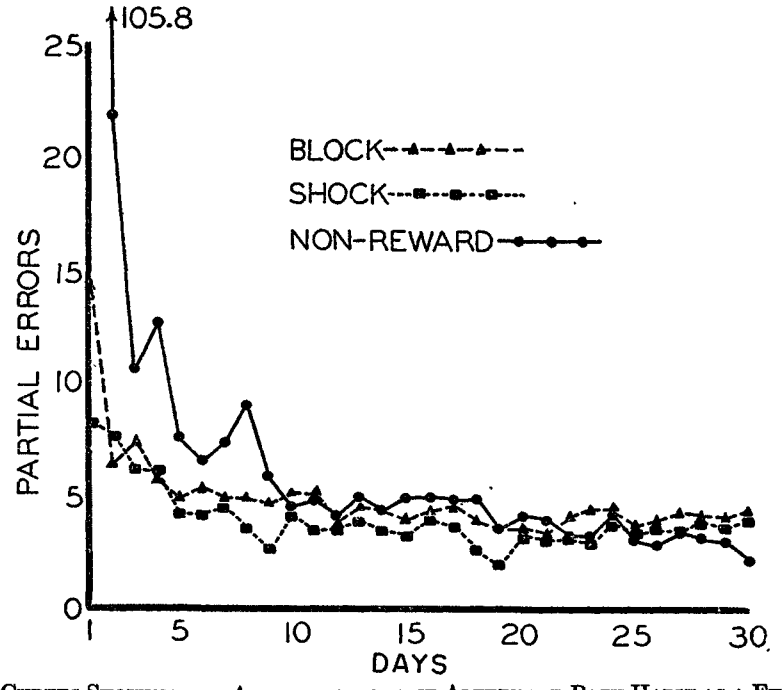


FIG. 7. CURVES SHOWING THE ACQUISITION OF THE ALTERNATE-PATH HABIT AS A FUNCTION OF THE TYPE OF REINFORCEMENT USED TO INHIBIT THE ORIGINAL-PATH HABIT

Each point represents the average number of "partial errors" made by each of the three groups of animals, during the Part-II training (cf. Fig. 5).

Although fairly elaborate time scores were kept for all animals in this part of the experiment, they proved to be much less useful than the error and partial-error scores in differentiating the performances of the various groups. For this reason we have restricted the discussion and analysis of this section to error and partial-error scores.

[9] Actually, the criterion of what constituted an "error" was the same (save in the event of conspicuous "interaction") for the long-, intermediate-, and short-path groups, since each of these groups was composed of five non-reward, five barrier, and five punishment animals. The differences in criteria of "error" which were correlated with the use of non-reward, barrier, and punishment thus tended to cancel out for the groups which were systematically varied only in so far as length of alternate path was concerned.

Part III. The results which have just been reported necessarily arouse certain expectations as to what the outcome of the third part of this experiment would be. Any set of conditions under which it is difficult to produce habit progression is likely to be correspondingly conducive to later regression, when the sanctions used in producing the progression are removed. From the previous description of the Part-III procedure, it will be recalled that in this part of the experiment all animals, after completing Part II, were run 15 trials a day for five days under the same conditions as prevailed in Part I. Under the Part-III conditions it was again possible for each animal to go from S to G, without let or hindrance, either by the alternate path which was available or by path O. Food was available at G regardless of the path taken from S to G.

This part of the experiment was designed to test the stability of the alternate-path habit, as a function of the various conditions of the experiment. The analysis consists of a comparison of the effects of differing lengths of the alternate path and of the different types of interference used to inhibit the original path-O habit on the tendency to revert to this habit when all "sanctions" were removed.[10]

The differing lengths of the alternate path should affect the stability of the alternate-path habit. The short-path group should show a more stable alternate habit than the intermediate-path group, and the alternate habit should be more firmly fixed in the intermediate group than in the long group. This proposition follows as a corollary from the earlier analysis in which it was shown that, due to the effects of time and effort, the alternate habit was strongest in the short path group, of medium strength in the intermediate group, and weakest in the long path group.

The empirical results confirm these expectations. Figure 8 shows that the short-path group regressed the least, the long-path group the most, and the intermediate-path group an intermediate amount.

Similarly, the shock group should regress less than either the non-reward or barrier groups. This follows from the fact that the alternate-path habit was reinforced by food, by escape from shock, and by escape from fear of shock in the former group but only by food in the two others; hence, the alternate habit should be strongest in the shock group and there should be the least amount of regression.

Figure 9 shows the differences in the tendency to regress as a function of the

[10] In previous laboratory studies it seems to have been the universal practice to induce regression by means of conflict resulting from the use of electric shock or some other form of "punishment" applied to the subject in connection with the performance of the more recently acquired habit. It will be noted that in the present experiment, we relied entirely upon the factor of effort to produce conflict and the resulting tendency toward regression. The fact that such a tendency is clearly manifested may be taken as further proof that there is a "strain toward inertia" in all behavior, *i.e.*, that there is necessarily present in every act an element of effort, or fatigue, which operates to prevent that act, no matter how satisfying, from becoming perfectly stereotyped, or "fixated". These considerations lead us toward the problem of behavior variability, which has been dealt with in a preliminary way elsewhere (14). They also show that the regression observed in Part III of this study was not, strictly speaking, "spontaneous".

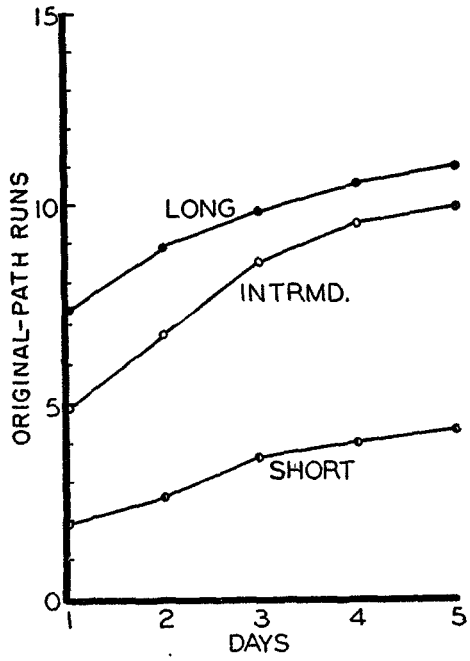


FIG. 8. CURVES SHOWING AMOUNT OF REGRESSION AS A FUNCTION OF THE LENGTH OF THE ALTERNATE-PATH DURING THE FIVE DAYS OF PART-III

Each point represents the average number of regressive (original-path) runs made by the 15 animals in each of these groups.

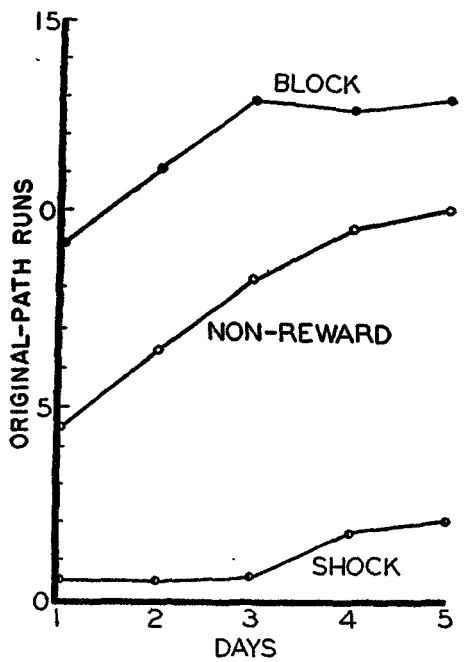


FIG. 9. CURVES SHOWING AMOUNT OF REGRESSION AS A FUNCTION OF THE TYPE OF INTERFERENCE DURING THE FIVE DAYS OF PART-III

Each point represents the average number of regressive runs made by the 15 animals in each of these groups.

method used in producing progression. It can be seen that the shock group regressed markedly less than the other two. That the non-reward group should have shown more resistance to regression than the barrier group is presumably related to the crossing of the curves representing these two groups in Figure 5. The explanation of this phenomenon is complicated and would take us beyond the scope of this paper.[11]

Part IV. Since the animals which had been shocked in Part II as a means of making them give up path O showed so little tendency to return to it "spontaneously" during Part III, it seemed desirable to run these animals, along with those of the non-reward and barrier groups, for 15 trials on a final day, with conditions such that the only way in which an animal could reach G was by taking path O.

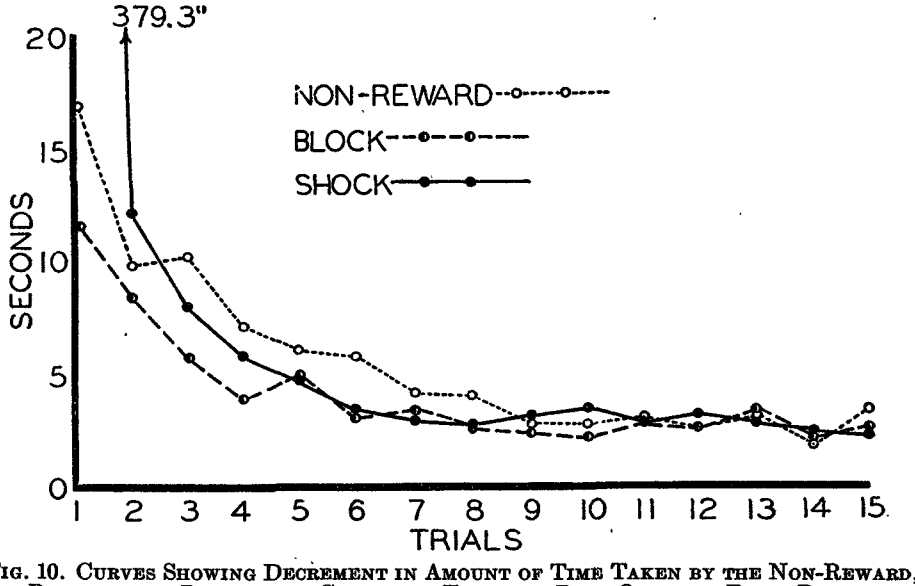


FIG. 10. CURVES SHOWING DECREMENT IN AMOUNT OF TIME TAKEN BY THE NON-REWARD, BARRIER, AND PUNISHMENT GROUPS TO TRAVERSE PATH O ON THE FINAL DAY OF THE EXPERIMENT (PART IV)

Each point represents the average number of seconds required by the 15 animals in each of these groups to carry out this act, with the alternate route from S to G blocked.

Since there was only one path now available as a route from S to G and since an animal could not make an "error" of any kind, the only remaining criterion of differentiation between groups was time. The curves presented in Figure 10 indicate that on the first trial under these particular conditions, it took the animals in the barrier group only 11.5″ on the average to go from S to G. The animals in the non-reward group took 16.9″ on the average to make this trip.

[11] Since the criterion of regression was the same for all three groups and since the results shown in Figure 9 are in agreement with the expectations generated by the curves shown in Figure 5, we may conclude that the difference in the criteria of "errors" in Part II is of no importance and does not invalidate the results shown in Figure 5. In fact, it would now appear, on the basis of the results shown in Figure 9, that "errors" are a more valid index of what happened in Part II than are "partial errors".

But the animals in the punishment group delayed 379.3″ on the average before they went from S to G via path O on the first trial. However, as soon as the latter animals had once been forced to "reality test" and had discovered that the shock was no longer present on path O, they made the trip from S to G on the remaining 14 trials as quickly as did the animals in the other two groups. This striking finding will be briefly commented on below.

DISCUSSION

The writers would be the last to maintain that the laboratory investigation which is here reported offers anything like a perfect paradigm of what happens in the socialization of human beings. In fact, one of us has elsewhere (16) commented at length upon the various ways in which this experiment does *not* parallel the education which both primitive and civilized children receive. However, we believe that in this study we have dealt with certain factors which are inevitably present and operative in the learning dilemmas that human beings have to pass through in order to "grow up" socially and intellectually. And we further believe that the results thus obtained have useful applications and implications if transferred, with due caution, to the human level.

In a sense, the various parts of the experiment parallel the various stages in the human life history. Part I, permitting as it does free access to the goal by the most direct route, parallels the indulgence of infancy—the time when the fewest restrictions are placed by society upon the satisfactions of the infant. Part II, involving interference with previously learned habits, parallels situations which frequently obtain during childhood, when previously learned habits are interfered with by various types of social sanctions and the more indirect habits demanded by society are substituted for them. Part III is more or less comparable to the "age of discretition", *i.e.*, the period during which the individual is "tested" by the withdrawal of many of the external social pressures which were in force in childhood but are supposed to be replaced by good character, or "conscience", in adult life. In Part IV the situation roughly parallels those human dilemmas in which established adult modes of action become realistically impossible or at least exceedingly difficult and the individual is faced by the choice between regression (or criminality) and serious deprivation or even death.

Our finding that the ease with which habit progression can be produced and the likelihood of later habit regression are both related to the effortfulness of the new adjustment squares well with common experience. This finding suggests the pedagogical maxim that when an original habit, or "fixation", must be interfered with, this end can be most readily accomplished, other things being equal, when a relatively easy alternative adjustment is available.[12] Whether, in circumstances in which there must ultimately be a very considerable discrepancy in the

[12] Incidental observations made in this study indicate that the longer the alternate path which the animals had to take to their goal, the more "frustrated" they appeared to be. The amount of gnawing on the maze and the number of attempts to "leave the field" (by crawling or jumping off the maze, etc.) were definitely greater in the long-path group than in the other two groups.

effortfulness of an earlier and a later habit, it is preferable to make the transition abruptly or in small steps is a question which is raised but not answered by our experiment. Our findings also raise certain interesting questions concerning the causes of behavior variability and provide some cues for later research.

The dramatic differences which resulted in this experiment from the use of different methods of interfering with the original habit and thereby producing habit progression again parallel, in a general way, what is known from the training of children. Punishment, such as that provided by the electric shock, is known to be maximally efficient in "breaking" habits, and our results indicate that their inhibitory effects are also most enduring. It is therefore easy to understand the widespread popularity of this method of "discipline". However, if flexibility and capacity to modify behavior with changing circumstances are also regarded as important, the use of punishment appears as a drawback rather than an advantage. As we have seen, those subjects with which shock had been used as a means of interfering with the original habit were slowest to "reality test" (3, 4) and discover that the original habit was again available to them. To the extent that they continued taking a more effortful route to their goal after the original route had become once more available to them, they may be said to have behaved "unrealistically".

As between the use of non-reward and a physical barrier as means of interfering with an original adjustment, it seems clear that, at least under the conditions of this experiment, non-reward is somewhat slower but more permanent in its effects.

The results obtained in Part IV raise the question as to whether anything resembling "conscience" was produced in the rats used in this study. It is true that when physically barred from taking the alternate path to their goal, the animals which had previously been shocked for taking the original path showed a great deal of hesitation, on the first trial, before reverting to the original path. However, as soon as they had done so *once*, they reverted immediately and completely to the original habit. In other words, as soon as they discovered that the shock was no longer present on path O, they promptly returned to the use of this path. They must thus be said to have had anxiety about the use of this path, but hardly "conscience". Conscience, at the human level, involves not only anxiety in the face of temptation but also guilt afterwards if the temptation is succumbed to. No evidence of guilt was to be seen in our rats. If, however, during the prior training the punishment for taking path O had been indeterminate, coming sometimes while the act was in progress, at other times occurring at varying intervals after the act had been completed, and on still other occasions perhaps not occurring at all, it seems not unlikely that the result obtained in Part IV would have been notably different.[13] It is perhaps too much to suggest that under these circumstances the rats might show something resembling the rudiments of "conscience" and "guilt"; but the experimental possibility thus suggested is an entertaining one. That further research along these lines might lead to a better understanding of the difficulties involved in psychotherapy is also worth noting.

[13] Cf. the results reported by Mowrer and Jones (15) and others on the greater resistance to extinction shown by habits established by intermittent than by constant reinforcement.

CONCLUSIONS

1. When rats are free to choose between two routes to the same goal, they develop a preference for the shorter of these routes.

2. This preference is not, however, an all-or-none affair but is a function of the magnitude of the difference in the length of the two routes.

3. Even when this difference is very great (a ratio of 8:1), preference for the shorter path is still not perfect.

4. The tendency for a certain amount of vacillation, or behavior variability, to continue despite very unequal reinforcement raises some fundamental questions which must be further investigated.

5. The ease with which rats can be forced to abandon the shorter of two routes to a goal and accept the longer as a substitute is inversely related to the discrepancy in the lengths of the two routes.

6. The ease with which rats can be forced to abandon the shorter of two routes to a goal and accept the longer as a substitute is also a function of the method used to interfere with the original habit, punishment (in the form of electric shock) being most effective, a physical barrier next most effective, and non-reward least effective.

7. Although, under the conditions of this experiment, a physical barrier showed an over-all superiority to non-reward as a method of producing habit progression, the non-reward method ultimately surpassed the barrier method.

8. Regardless of the method used to produce habit progression, rats show a tendency in this type of experiment to persist in behavior variability which serves to determine whether the penalty for taking the originally preferred route is "still there".

9. When the "sanctions" used to enforce habit progression are removed, the tendency to revert to the original adjustment is positively related to the degree of difficulty encountered in producing the habit progression.

10. Under the conditions of this experiment, the regression which occurs when "sanctions" are removed is never complete, being inversely related to the level of habit progression which has previously been attained.

11. When regression to the original path is forced, the punishment animals show much greater reluctance to revert to the original adjustment than do the barrier and non-reward animals.

12. The results of this investigation have certain implications for education, mental hygiene, and psychotherapy.

REFERENCES

1. Benedict: Continuities and discontinuities in cultural conditioning. Psychiatry, 1938, **1**, 161–167.
2. Fisher, R. A.: The Design of Experiments. Oliver and Boyd, London: 1937.
3. French, T.: Reality Testing in Dreams. Psychoanalytic Quarterly, 1937, **6**, 62–77.
4. Freud, S.: Metapsychological supplement to the theory of dreams (1916). Collected Papers, Vol. IV, pp. 137–151. Hogarth Press, London: 1937.
5. Hamilton, J. A., and Krechevsky, I.: Studies in the effect of shock upon behavior plasticity in the rat. J. Comp. Psychol., 1933, **16**, 237–253.
6. Hull, C. L.: The concept of habit-family hierarchy and maze learning. Psychol. Rev., 1934, **41**, 33–54 and 134–152.

7. HULL, C. L.: Principles of Behavior. D. Appleton-Century, New York: 1943.
8. KLEEMEIER, R. W.: Fixation and regression in the rat. Psychol. Monog., 1942, **54**, pp. 34.
9. LINTON, R.: Age and sex categories. Am. Soc. Review, 1942, **7**, 589–603.
10. MARTIN, R. F.: 'Native' traits and regression in the rat. J. Comp. Psychol., 1940, **30**, 1–16.
11. MEAD, M.: Growing up in New Guinea: Morrow, New York, 1930.
12. MOWRER, O. H.: An experimental analogue of 'regression' with incidental observations on 'reaction-formation'. J. Ab. and Soc. Psychol., 1940, **35**, 56–87.
13. MOWRER, O. H.: Motivation and learning in relation to the national emergency. Psychol. Bull., 1941, **38**, 421–431.
14. MOWRER, O. H., AND JONES, HELEN M.: Extinction and behavior variability as functions of effortfulness of task. J. Exp. Psychol. (in press).
15. MOWRER, O. H., AND JONES, HELEN M.: Habit strength as a function of pattern of reinforcement (in preparation).
16. MOWRER, O. H., AND KLUCKHOHN, C.: Dynamic theory of personality, Chapter III. *In* Personality and the Behavior Disorders. Edited by J. McV. Hunt. Ronald Press: New York, 1943.
17. O'KELLY, L. I.: An experimental study of regression. I. Behavioral characteristics of the regressive response. J. Comp. Psychol., 1940, **30**, 41–53.
18. O'KELLY, L. I.: An experimental study of regression. II. Some motivational determinants of regression and perseveration. J. Comp. Psychol., 1940, **30**, 55–95.
19. PARSONS, .: Age and sex in the social structure of the United States. Am. Soc. Review, 1942, **7**, 604–620.
20. PORTER, E. H., AND BIEL, W. C.: Alleged regressive behaviour in a two-unit maze. J. Comp. Psychol., 1943, **35**, 187–195.
21. SANDERS, M. J.: An experimental demonstration of regression in the rat. J. Exp. Psychol., 1937, **21**, 493–510.
22. STECKLE, D. C., AND O'KELLY, L. I.: The effect of electrical shock upon later learning and regression in the rat. J. Psychol., 1940, **9**, 365–370.
23. WHEELER, R. H.: The Science of Psychology. New York: Crowell, 1941.
24. WHITING, .: Becoming a Kwoma. Yale Univ. Press, New Haven, 1941.
25. YOSHIOKA, J. G.: Weber's law in the discrimination of maze distance by the white rat. Univ. Calif. Publ. Psychol., 1929, **4**, 155–184.

APPENDIX

The two principal advantages which accrue from the use of a factorial design in psychological experimentation are: (1) it is more economic of subjects and, therefore, of the experimentor's time; and (2) it allows the use of analysis of variance (*F*-test), a statistical technique which, among other distinctions, uniquely provides for the precise measurement of "interaction" between two or more independent variables. The first of these advantages is illustrated by the fact that, by using a factorial design, it was possible in the present study to obtain from 45 subjects essentially the same (in some respects more) information than would have been obtained from two separate experiments of conventional design involving *twice* the number of subjects here employed. But in the present instance a special difficulty was encountered in applying the analysis of variance technique. Since this type of difficulty arises relatively frequently in psychological research, it warrants a somewhat extended discussion. The following quotations will indicate the source of this difficulty.

"It is important to note that the evaluation of individual group means and the individual differences between them involves the assumption that the variance within a group is the same, except for chance, from group to group. In other words, we assume that whatever factor has resulted in significant differences in group *means* will *not* also result in significant differences in group variances. If the group variances do differ fundamentally, then, of course, the standard error of the mean will differ from group to group, even though all groups are of the same size, and it would not be valid to compute the standard error of a mean from the 'variance *within groups*.' This assumption, known as the assumption of *homogeneity of variance*, may not be very well satisfied in the typical methods experiment. . . ."[1]

"*Correlation between mean and variance.* A fundamental condition for the validity of the *F*-test of significance is that the two mean squares be independent. If one is conditioned by the other in any manner, the test is invalid. The popularity of analysis of variance has resulted in its application to data in which this condition is not met. . . . This decreases the reliability of the test of significance."[2]

Inspection of Figure 1A will show that in the case of the results obtained in Part II of the present study, there is a *perfect* rank-order correlation between the average number of "errors" made by the animals in the nine sub-groups and the variability (σ) shown by the five animals constituting each of these sub-groups. In other words, as the mastering of the required learning task (progression to the alternate route to the goal) became increasingly difficult, there was a tendency for the subjects to manifest progressively larger "individual differences". This type of correlation has been repeatedly noted previously[3] and need not be further considered here except to say that it flagrantly violates the assumption of "homogeneity of variance" and makes the analysis of variance inapplicable, at least as far as the original "error" data are concerned.

In an attempt to circumvent this difficulty, the "error" scores shown in Figure 1A were converted into their natural logarithms.[4] When this was done, the rank-order correlation between mean scores and variability was changed from $+1.0$ to -0.4, a value sufficiently near the ideal of zero correlation to make possible an analysis of variance with results which are at least reasonably valid. These results are given in Figure 2A[5] and indicate that the differences due to methods used to interfere with the original-path habit, the differences due to the length of the alternate path, and the interaction between these two sets of variables were all significant at the 0.01 level or better. In non-statistical language, the phenomenon of interaction here means that, as can be seen in Figure 1A, the tendency for the number of "errors" to increase as the length of the alternate path is increased is greater in the non-reward groups than in the barrier and punishment groups. At present we have no explanation to offer as to why this should have been the case.

[1] Lindquist, E. F.: Statistical analysis in educational research. Houghton Mifflin: New York, 1940, p. 99.

[2] Snedecor, G. W.: Statistical Methods. Collegiate Press: Ames, Iowa, 1938, p. 207.

[3] Mowrer, O. H., and Lamoreaux, R. R.: Avoidance conditioning and signal duration—A study of secondary motivation and reward. Psychol. Monog., 1942, **54**, 1–34.

[4] Snedecor, *ibid*, p. 207.

[5] The writers are indebted to Miss Helen Jones and to Mr. Dalton Vernon for aid in making these and certain other statistical computations reported in this paper.

As was intimated in the main body of this paper, the design of the experiment here reported is somewhat short of ideal in that there was "confounding" of the differences (if any) caused by the particular manner in which the five experimenters worked and the differences (if any) caused by the sex of the rats. The

LENGTH OF ALTERNATE PATH	METHOD OF INTERFERING WITH ORIGINAL HABIT		
	Non-reward	Barrier	Punishment
Short	(E1) 67 (E2) 196 M = 92.0 (4) (E3) 51 σ = 47.6 (4) (E4) 83 (E5) 63	(E1) 28 (E2) 65 M = 44.6 (6) (E3) 36 σ = 13.7 (6) (E4) 38 (E5) 56	(E1) 2 (E2) 3 M = 4.2 (9) (E3) 2 σ = 2.7 (9) (E4) 8 (E5) 6
Interm.	(E1) 117 (E2) 266 M = 193.8 (2) (E3) 180 σ = 58.1 (2) (E4) 255 (E5) 151	(E1) 41 (E2) 63 M = 81.6 (5) (E3) 170 σ = 46.6 (5) (E4) 49 (E5) 85	(E1) 2 (E2) 10 M = 8.2 (7) (E3) 5 σ = 6.9 (7) (E4) 3 (E5) 21
Long	(E1) 748 (E2) 471 M = 405.2 (1) (E4) 402 σ = 202.4 (1) (E4) 182 (E5) 223	(E1) 162 (E2) 106 M = 118.4 (3) (E3) 201 σ = 55.4 (3) (E4) 60 (E5) 63	(E1) 4 (E2) 5 M = 5.8 (8) (E3) 4 σ = 3.6 (8) (E4) 3 (E5) 13

FIG. 1A. TABLE SHOWING TOTAL NUMBER OF "ERRORS" MADE BY EACH OF 45 ANIMALS DURING THE 30 DAYS OF TRAINING CONSTITUTING PART II OF THIS EXPERIMENT

E1 stands for Experimenter 1; E2, Experimenter 2; etc. The numbers in parentheses a the right of the mean (M) and standard deviation (σ) values represent the rank orders of these values.

SOURCES OF VARIATION	DEGREES OF FREEDOM	SUM OF SQUARES	VARIANCE	"F"	LEVEL OF SIGNIFICANCE
Methods used to interfere with original habit	2	7.61527	3.80763	59.6433	5.25 = 0.01
Lengths of alternate path to goal	2	1.21278	0.60639	9.4979	5.25 = 0.01
Interaction	4	12.73040	3.24368	50.8063	3.89 = 0.01
Within groups	36	2.29840	0.06384		
Total	44	23.85685			

FIG. 2A. TABLE SHOWING RESULTS OF ANALYSIS OF VARIANCE CARRIED OUT IN MANNER DESCRIBED IN TEXT

rats used by experimenters E1, E3, E4, and E5 were all male, but those used by experimenter E2 were female. It is consequently impossible to say whether any deviation in the results obtained by E2 from those obtained by the other experimenters is due to the sex of E2's subjects or to his distinctive behavior in conducting his part of the investigation. If the rats had been either all male or

all female or if the two sexes had been *randomly* distributed among the five experimenters, analysis of variance would have provided an exact measure of the differences introduced both by the sex of the subjects and by the idiosyncracies of the various experimenters[6]. However, since these two factors were confounded, all differences due to either or both of them had to be classed as "error" (Variation within Groups, See Figure 2A) and lumped with those unaccountable variations which occurred between animals receiving the same kind of experimental treatment. If this procedure had any effect, it would necessarily be that of making the *F*-values recorded in Figure 2A spuriously small. In other words, if this experiment had been ideally designed, so that the measure of "error", would not have included the variability introduced by the differences between experimenters and between the sex of the subjects, the levels of significance recorded in Figure 2A would have been at least as high as and quite possibly higher than those actually obtained. Thus, the fault in the experimental design of this investigation may have depressed the significance of the obtained results, but it is unlikely that they had the reverse effect. Since the results seem trustworthy, despite this defect, it is probably of no importance.

As a matter of fact, it is possible to show that the differences between experimenters and between the sex of the rats were almost certainly *not* significant in determining the results presented in Figure 1A, i.e., did not depress the *F*-values shown in Figure 2A. In Figure 3A the results presented earlier in Figure 1A have been rearranged according to experimenters. Although there is considerable range (from 1185 to 681) in the total number of "errors"[7] made by the nine animals which were run by each of the five experimenters, none of the differences is statistically significant. Thus, the *t*-value for the difference between the results obtained by E1 and E5 is 0.33 and that for the difference between the results obtained by E2 and E4 is likewise only 0.59 (which means that the corresponding *P*-values are roughly 0.70 and 0.60). From these findings it would again appear that the confounding (of experimenters and sex of subject) which occurred in this experiment, while objectionable in principle, made little or no practical difference in the results actually obtained.

Quite aside from the problem of "confounding" which has just been considered, the results of the part of the present investigation which are under discussion here do not, because of lack of "homogeneity of variance", lend themselves particularly well to the analysis of variance technique. Since this same difficulty is encountered relatively frequently in psychological data, the question arises as to

[6] Since each experimenter ran only one rat under each of the nine experimental conditions, the variance "between rats" and "between experimenters" were necessarily confounded. Only by (1) having each experimenter run at least two rats under each experimental condition or (2) resorting to the dubious expedient of using "higher order interactions" (cf. Crutchfield, R. S. Efficient factorial design and analysis of variance illustrated in psychological experimentation. *J. Psychol*, 1938, **5**, 339-346) could these two sources of variance have been independently estimated.

[7] The reader should remember that the term "error" as employed in this context has a meaning entirely different from that of the statistical term "error" (normally, variability between subjects which are run under the same experimental conditions).

just how suitable factorially designed experiments are in this field.[8] It is true that analysis of variance is essential to the determination of "interaction", but the inapplicability of this technique does not otherwise greatly impair the effectiveness of a factorially designed experiment. Fortunately, the "homogeneity of variance", which is a prerequisite for the valid application of the *F*-test, is not a desideratum in the use of the *t*-test. Although only the former can be used to determine the over-all significance of differences between *three* or more experimental groups, the latter is often ideally suited for determining the reliability of differences between *two* groups. Thus, it is entirely legitimate, for example, to use the *t*-test as a means of determining the reliability of the differences between the Non-reward and Barrier groups, the Barrier and Punishment groups, and the Non-reward and Punishment groups and of the differences between the Short and Intermediate groups, the Intermediate and Long groups, and the Short and Long groups (see Figure 1A). Applying the *t*-test in the conven-

	E1	E2	E3	E4	E5
S-NR	67	196	51	83	63
S-B	28	65	36	38	56
S-P	2	3	2	8	6
I-NR	117	266	180	255	151
I-B	41	63	170	49	85
I-P	2	10	5	3	21
L-NR	748	471	402	182	223
L-B	162	106	201	60	63
L-P	4	5	4	3	13
Total	1171	1185	1051	681	681

FIG. 3A. DATA FROM FIG. 1A REARRANGED ACCORDING TO EXPERIMENTERS (E1, E2, ETC.)

The notations at the left refer to the nine experimental conditions used in this investigation (see Fig. 1 and Fig. 1A).

tional manner described by Fisher,[9] values of 3.70, 5.59, and 3.43 and of 3.35, 1.78 and 2.74 were obtained, respectively, for the differences between six pairs of groups just listed. With 14 degrees of freedom (number of subjects in each group minus one), a *t*-value of 1.76 is significant at the 0.10 level, a value of 2.15

[8] Garrett and Zubin (The analysis of variance in psychological research. Psychol. Bull., 1943, **40**, 233–267) touch upon this problem but do not follow up its implications. (Cf. also Baxter, B. The application of factorial design to a psychological problem. Psychol. Rev., 1940, **47**, 494–500.)

[9] Fisher, R. E.: The design of experiments. Oliver and Boyd: Edinburgh, 1937, pp. 38–43. For a discussion of the *t*-test as applied in situations where items are "paired" (as in the present instance) and where they are not, see Lindquist, *ibid.* Cf. also the discussion by Garrett (H. E.: Statistics in Psychology and education. Longmans, Green & Co.: New York, 1939) of the use of the "critical ratio" technique where the two series under examination are or are not "correlated". When items are "paired", or "correlated", the *t*-test procedure is notably simpler and more precise.

at the 0.05 level, a value of 2.62 at the 0.02 level, and a value of 2.98 at the 0.01 level. With one exception, all of the obtained t-values thus reach or surpass the conventional (0.05) criterion of reliability.[10]

Statistical analyses of the results obtained in Parts III and IV of this investigation have not been made, but the curves shown in Figures 8, 9, and 10 differ so dramatically that there can be little doubt of the significance of the observed discrepancies.

[10] It should be kept in mind that the t-values given above are based on the data presented in Figure 1A, which consist of the total number of "errors" made by each of the 45 animals during the entire 30-day period constituting Part II of this investigation. By referring back to Figure 4 and Figure 5, it will be seen that the differentiation between groups is most marked in the early stages of this period and becomes less so toward the end. Therefore, in using the number of "errors" made by each animal throughout this entire period as the basis of group comparisons, the differences between groups are necessarily less striking and less reliable than they would be if the comparisons were based on the "error" scores made, let us say, on the first day or during the first 10 days. It goes without saying that a t-test could be applied at any point in the whole 30-day procedure, either to the cumulative scores up to that date or to the scores representing the performance of the various groups *on* that particular date. However, for general purposes the total "error" scores seem the most meaningful and have accordingly been used for the foregoing analyses. (The fact that all of the curves shown in Figure 4 and Figure 5 tend to level off instead of converging on the baseline representing no "errors" is related to a tendency noted in differing degrees in all groups to persist, seemingly indefinitely, in a certain amount of variable, or exploratory behavior. This phenomenon of continued "reality testing" has interesting implications for the understanding of what Krechevsky has called "behavior plasticity" which will be elaborated at a later time.)